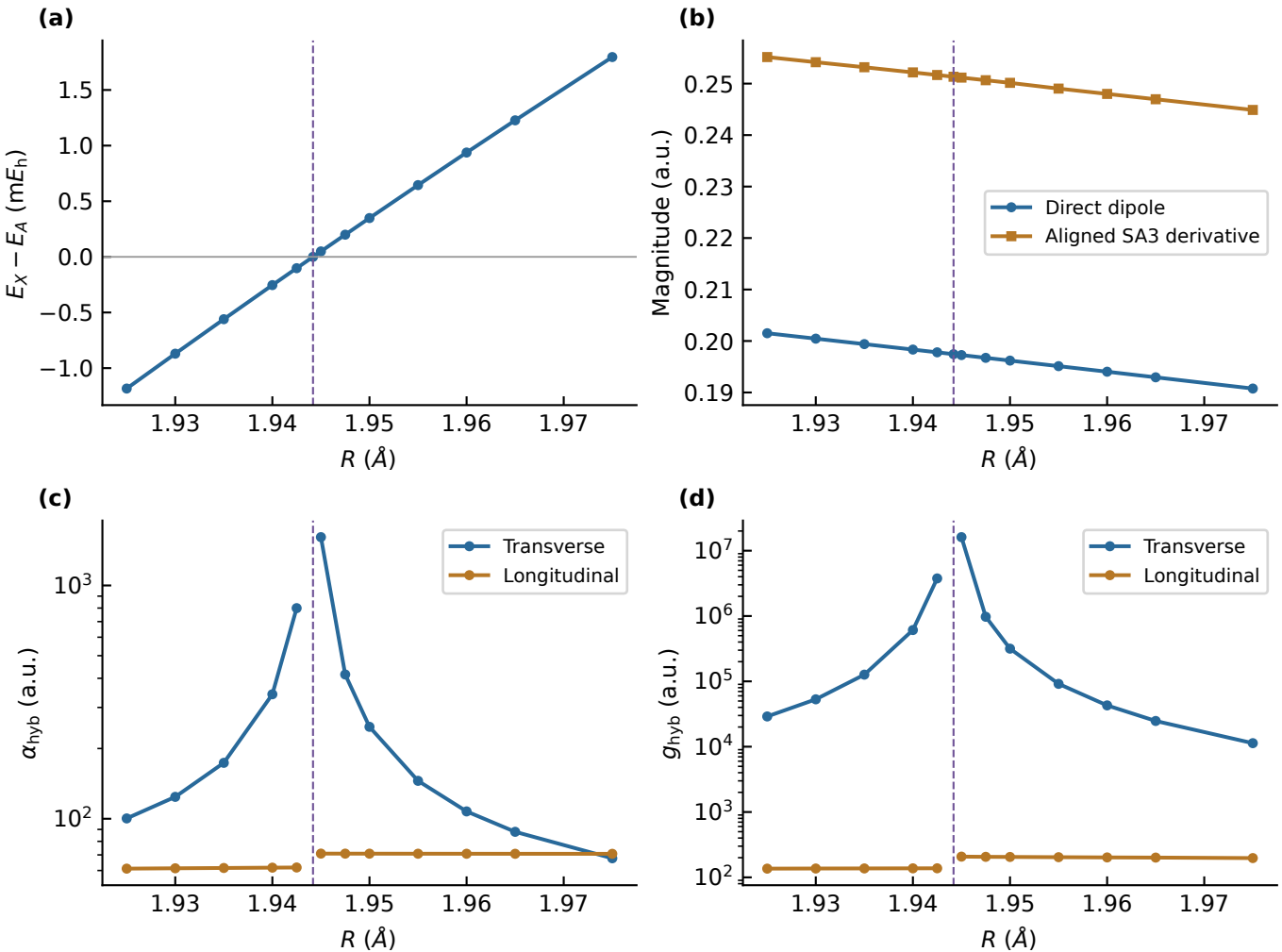
(a)
$E_X - E_A$ (mE_h)
1.5
1.0
0.5
0.0
−0.5
−1.0
1.93
1.94
1.95
1.96
1.97
R (Å)
(b)
Magnitude (a.u.)
0.25
0.24
0.23
0.22
0.21
0.20
0.19
Direct dipole
Aligned SA3 derivative
1.93
1.94
1.95
1.96
1.97
R (Å)
(c)
α_{hyb} (a.u.)
10^3
10^2
Transverse
Longitudinal
1.93
1.94
1.95
1.96
1.97
R (Å)
(d)
g_{hyb} (a.u.)
10^7
10^6
10^5
10^4
10^3
10^2
Transverse
Longitudinal
1.93
1.94
1.95
1.96
1.97
R (Å)